

Experiment No. 8

Carrier Phase Synchronization using Costas Loop

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Aim

To study carrier phase synchronization using a Costas Loop in a QPSK communication system implemented using GNU Radio and PlutoSDR.

Objective

The objective of this experiment is to recover carrier phase information from a received QPSK signal and achieve accurate constellation synchronization using a Costas Loop.

Theory

In digital communication systems, carrier phase offsets occur due to oscillator mismatches, channel impairments, and propagation effects. These phase errors rotate constellation points and increase symbol detection errors.

Carrier Phase Synchronization is the process of estimating and correcting phase errors between the transmitter and receiver carriers.

A Costas Loop is a specialized Phase Locked Loop (PLL) widely used for carrier recovery in QPSK and BPSK systems. It continuously estimates the phase error and adjusts the local oscillator to align the received carrier with the transmitted carrier.

The synchronization chain used in this experiment is:

Matched Filter $\rightarrow$ Symbol Timing Recovery $\rightarrow$ Costas Loop $\rightarrow$ Constellation Recovery

The Costas Loop performs fine carrier synchronization after frequency and timing synchronization have been completed.

Equipment Required

- ADALM-PLUTO SDR
- GNU Radio Companion (GRC)
- Antenna
- USB Cable
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- libiio Support

Carrier Phase Synchronization System

The above flowgraph demonstrates carrier phase synchronization in a QPSK receiver. The Costas Loop block estimates and corrects carrier phase errors, resulting in stable constellation points and reliable symbol recovery.

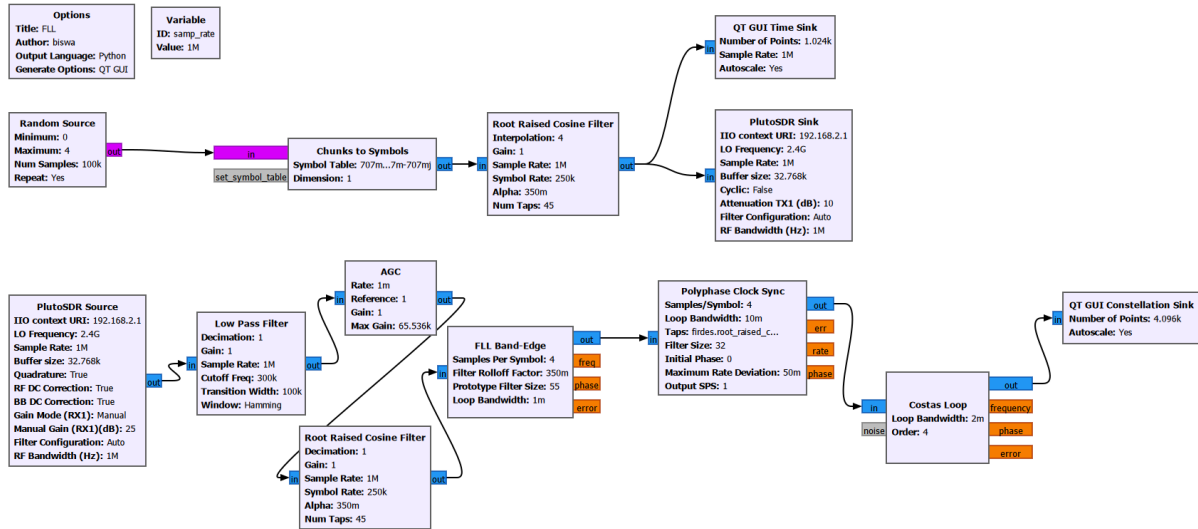


Figure 1: Carrier Phase Synchronization using Costas Loop

Flowgraph Description

PlutoSDR Source

Receives the transmitted QPSK signal.

Low Pass Filter

Removes unwanted frequency components and channel noise.

Automatic Gain Control (AGC)

Normalizes the signal amplitude before synchronization.

Root Raised Cosine Filter

Acts as a matched filter and improves signal quality.

Polyphase Clock Sync

Performs symbol timing recovery.

Costas Loop

Performs carrier phase synchronization by estimating and correcting phase offsets in the received signal.

QT GUI Constellation Sink

Displays the recovered constellation after synchronization.

Working Principle

The received QPSK signal is first filtered and normalized. Symbol timing recovery is then performed using the Polyphase Clock Sync block.

After timing synchronization, the Costas Loop estimates the carrier phase error and continuously adjusts the local oscillator phase. This process removes constellation rotation and stabilizes the received symbols.

The synchronized constellation can then be used for reliable symbol decoding.

Parameters

Parameter	Value
Sample Rate	1 MS/s
Symbol Rate	250 kSymbols/s
Samples per Symbol	4
Roll-off Factor (α)	0.35
Carrier Frequency	2.4 GHz
Costas Loop Order	4 (QPSK)
Loop Bandwidth	0.0628
Polyphase Filter Size	32
Output Samples per Symbol	1

Table 1: Carrier Phase Synchronization Parameters

Procedure

1. Open GNU Radio Companion.
2. Create the QPSK transmitter flowgraph.
3. Configure PlutoSDR Sink for transmission.
4. Create the receiver flowgraph using PlutoSDR Source.
5. Add Low Pass Filter and AGC blocks.
6. Add a Root Raised Cosine matched filter.
7. Configure the Polyphase Clock Sync block.
8. Add a Costas Loop block for carrier recovery.
9. Connect a QT GUI Constellation Sink.
10. Run the flowgraph.
11. Observe constellation behavior before and after synchronization.
12. Analyze carrier phase correction using the Costas Loop.

Observation

The Costas Loop successfully corrected carrier phase offsets present in the received signal. The constellation points became stable and properly aligned, resulting in improved symbol detection performance.

Result

Carrier phase synchronization was successfully achieved using the Costas Loop algorithm. The phase error was corrected, enabling stable constellation recovery and accurate symbol decoding.

Applications

- QPSK Communication Systems
- BPSK Communication Systems
- Satellite Communication
- Wireless Communication Systems
- Software Defined Radio Receivers
- Digital Modems

Conclusion

The experiment demonstrated carrier phase synchronization using a Costas Loop in a QPSK communication system. The Costas Loop effectively corrected phase offsets and improved receiver performance by enabling reliable constellation recovery and symbol detection.

References

1. GNU Radio Documentation
2. Analog Devices PlutoSDR Documentation
3. Digital Communication Systems by Simon Haykin
4. Software Defined Radio Principles and Applications